

國立暨南國際大學資訊工程學系博士班必選修科目一覽表

Doctoral Program Course Catalog of Department of Computer Science and Information Engineering, National Chi Nan University

本表(**110-114**)學年度入學新生適用 Effective for students admitted in Fall, **2021-2025**

最低畢業學分：18學分，逕行修讀博士學位者須修畢本系碩士班以上課程至少30學分。必修課程：專題研究（2學分）、專題討論（2學分）。以上不含博士論文（通過學位考試後給12學分）、專題研究、與專題討論之學分。

Doctoral program requirements: a minimum of 18 required credits plus the PhD dissertation (12 credits), and the extra credits earned from research projects and research seminars. Students transferred from other programs of the Department (e.g. the master program) to the doctoral program must complete at least 30 credits from the Department's master program.

課程代號 Course Code	課程名稱(中文及英文) Course Name(Chinese and English)	學分數 Credits	修別 Course type (required/elective)	開課年級 Course availability	備註 Notes
219001	專題研究（一） Independent Study (1)	1	必 (R)	一上 Doctorate 1st Semester	
219003	專題研究（二） Independent Study (2)	1	必 (R)	一下 Doctorate 2nd Semester	
219002	專題討論（一） Seminar (1)	1	必 (R)	一上 Doctorate 1st Semester	
219004	專題討論（二） Seminar (2)	1	必 (R)	一下 Doctorate 2nd Semester	
219102	資訊擷取 Data Mining and Information Extraction	3	選 (E)	一 Doctorate	
219094	智慧型行動裝置軟體設計 Intelligent Mobile Device Programming	3	選 (E)	一 Doctorate	
219014	計算複雜度 Computational Complexity	3	選 (E)	一 Doctorate	
219016	整合式訊源/通道編解碼設計（一） Joint Source-Channel Coding (I)	3	選 (E)	一 Doctorate	
219063	樣型識別 Pattern Recognition	3	選 (E)	一 Doctorate	
219064	整合式訊源/通道編解碼設計（二） Joint Source-Channel Coding (II)	3	選 (E)	一 Doctorate	
219067	智慧型系統實務（一） Intelligent System Practice (1)	3	選 (E)	一 Doctorate	

219068	非擬真成像技術 Non Photo-realistic Rendering	3	選 (E)	— Doctorate	先修電腦圖學及高等電圖 prerequisite: Computer Graphics & Advanced Computer Graphics
219088	量子資訊科學導論 Introduction to Quantum Information Science	3	選 (E)	— Doctorate	
219069	多媒體資訊分析與檢索 Multimedia Analysis and Indexing	3	選 (E)	— Doctorate	
219074	通道編碼與應用 Applications of Channel Coding	3	選 (E)	— Doctorate	
219079	Web知識擷取 Web Knowledge Extraction	3	選 (E)	— Doctorate	
219085	科技英文(一) Technical English(I)	3	選 (E)	— Doctorate	
219086	科技英文(二) Technical English(II)	3	選 (E)	— Doctorate	
219089	高等量子演算法 Advanced Quantum Algorithms	3	選 (E)	— Doctorate	
219090	量子計算與通訊 Quantum Computing and Communication	3	選 (E)	— Doctorate	
219093	智慧型系統實務(一) Intelligent System Practice (1)	2	選 (E)	— Doctorate	
219095	行動與雲端運算應用 Mobile Cloud Computing Applications	3	選 (E)	— Doctorate	
219097	進階Linux系統管理 Advanced Linux System Administration	3	選 (E)	— Doctorate	
219100	行動軟體系統設計 Deesign of Mobile Software System	3	選 (E)	— Doctorate	
219101	語言學習與科技 Language Learning and Technology	3	選 (E)	— Doctorate	
219103	機器學習 Machine Learning	3	選 (E)	— Doctorate	
219105	行動資訊程式設計 Mobile Information Programming	3	選 (E)	— Doctorate	
219108	行動式3D電腦繪圖 Mobile 3D Computer Graphics	3	選 (E)	— Doctorate	需先修電腦圖學及高等電腦圖學 prerequisite: Computer Graphics & Advanced Computer Graphics
219111	智慧型機器人程式設計 NXT Programming	3	選 (E)	— Doctorate	
219112	3D建模與互動多媒體應用 3D Modeling and Interactive Multimedia Applications	3	選 (E)	— Doctorate	
219113	網際網路電話加值服務 IP Telephony Service Creation Technology	3	選 (E)	— Doctorate	
219129	網際網路電話 IP Telephony	3	選 (E)	— Doctorate	
219126	網際網路通訊協定第六版 (IPv6) Internet Protocol Version 6	3	選 (E)	— Doctorate	
219120	網路電話品質量測技術 VoIP Quality Measurement	3	選 (E)	— Doctorate	

219119	計算機網路特論 Special Topics on Computer Networks	3	選 (E)	— Doctorate	
219070	智慧型系統實務（二） Intelligent System Practice (2)	2	選 (E)	— Doctorate	
219076	智慧型系統實務（三） Intelligent System Practice (3)	2	選 (E)	— Doctorate	
219080	智慧型系統實務（四） Intelligent System Practice (4)	2	選 (E)	— Doctorate	
219092	電子電路設計自動化 VLSI Electronic Design Automation	3	選 (E)	— Doctorate	
219099	雲端系統設計 Design of Cloud Computing System	3	選 (E)	— Doctorate	
219114	人工智慧程式交易系統 Automated Trading System using Artificial Intelligence	3	選 (E)	— Doctorate	
219115	音樂,大腦,與科技 Music,Brain,Science and	3	選 (E)	— Doctorate	
219116	英文寫作, 口說, 簡報 (一) On English Writing, Speaking and Presentation (1)	3	選 (E)	— Doctorate	
219117	英文寫作, 口說, 簡報 (二) On English Writing, Speaking and Presentation (2)	3	選 (E)	— Doctorate	
219091	智慧型演化計算 Intelligent Evolutionary	3	選 (E)	— Doctorate	
219121	尖端資訊科技國際課程 International Studies on Leading Information Technologies	1	選 (E)	— Doctorate	
219130	安卓程式設計 Android Programming	3	選 (E)	— Doctorate	
219122	QNAP應用程式設計 Linux Programming for QNAP	3	選 (E)	— Doctorate	
219123	開源電子原型平台軟硬體整合設計與實作 software / hardware co-design & implementation for open-source electronics prototyping platform	3	選 (E)	— Doctorate	
219125	高等資料壓縮 Advanced Data Compression	3	選 (E)	— Doctorate	
219127	ISMS資訊安全認證實務 Practice in Information Security Management System Certification	3	選 (E)	— Doctorate	
219128	數位音樂學習與創作 Digital Music Learning and Creation	3	選 (E)	— Doctorate	
219133	國際資訊科技發展趨勢 Global Trend of Information Technology Development	2	選 (E)	— Doctorate	
219134	尖端資訊科技國際課程（二） International Studies on Leading Information Technologies(2)	2	選 (E)	— Doctorate	

219132	研究所企業實習 Graduate Enterprise Internship	3	選 (E)	— Doctorate	
219131	軟體定義網路 Software-Defined Network	3	選 (E)	— Doctorate	
219135	交響樂團與數位應用 Orchestra and Its Digital	2	選 (E)	— Doctorate	
219136	音樂科技專題策展 Capstone Project on Music Technology	2	選 (E)	— Doctorate	
219138	音樂, 大腦, 與科技 Music, Brain, Science and Technology	2	選 (E)	— Doctorate	
219139	數位音樂學習與創作 Digital Music Learning and Creation	2	選 (E)	— Doctorate	
219140	PHP 網頁程式設計 PHP Web Programming	3	選 (E)	— Doctorate	
219141	計算智能 Computational Intelligence	3	選 (E)	— Doctorate	
219142	大數據系統實務 Big Data System and Practice	3	選 (E)	— Doctorate	
219143	深度學習 Deep Learning	3	選 (E)	— Doctorate	
219144	Python網管程式設計 Python for Network Management and System Administration	3	選 (E)	— Doctorate	
219145	高等計算智能 Advanced Computational	3	選 (E)	— Doctorate	需先修計算智能 prerequisite: Computational Intelligence
219146	程式練習及訓練與競賽課程 Programming Practice, Training, and Contest Course	3	選 (E)	— Doctorate	
219147	計算機網路特論二 Special Topics on Computer Networks (II)	3	選 (E)	— Doctorate	需先修計算機網路特論 prerequisite: Special Topics on Computer Networks
219148	SQL程式設計 SQL Programming	3	選 (E)	— Doctorate	
219149	程式練習及訓練與競賽(二) Programming Practice, Training, and Contest Course (II)	3	選 (E)	— Doctorate	
219150	前端程式設計 Frontend Programming	3	選 (E)	— Doctorate	
219151	IT科技日文 IT/Technical Japanese	3	選 (E)	— Doctorate	需先修日文一(上)、日文一(下) prerequisite: Japanese I(1)、Japanese I(2)
219152	Python網頁擷取程式設計 Web Scraping with Python	3	選 (E)	— Doctorate	需先修計算機概論 prerequisite: Introduction to Computer Science
219153	量子啟發式搜尋演算法 Quantum-inspired heuristic and evolutionary algorithms	3	選 (E)	— Doctorate	
219154	命名資料網路 Named Data Networking	3	選 (E)	— Doctorate	需先修計算機網路 prerequisite: Computer Networks

219155	Python機器學習實務 Machine Learning Practice with Python	3	選 (E)	— Doctorate	
219156	iOS App 程式開發 iOS App Programming	3	選 (E)	— Doctorate	
219157	FPGA實作深度學習方法 FPGA Implementation of Deep Learning Methods	3	選 (E)	— Doctorate	需先修邏輯設計或邏輯設計與實驗 prerequisite: Logic Design or Logic Design and Laboratory Experiments
219158	Python 網路程式設計 Python Network Programming	3	選 (E)	— Doctorate	需先修計算機網路 prerequisite: Computer Networks
219159	情境模擬控制系統之軟硬體整合 應用實務 Hardware / Software Integration & Practice for Situational Simulation Control System	3	選 (E)	— Doctorate	
219160	智慧型積木機器人程式設計實務 EV3 Intelligent Lego Robot Programming	3	選 (E)	— Doctorate	
219161	音樂程式設計 Music Programming	3	選 (E)	— Doctorate	
219162	Python特洛伊木馬程式設計 Design Trojan Horses with Python Language	3	選 (E)	— Doctorate	需先修Python網路程式設計 prerequisite: Python Network Programming
219163	全球資訊新聞導讀 Global Information Technology News	3	選 (E)	— Doctorate	
219164	5G通訊架構之網路切片技術 Network Slicing in 5G	3	選 (E)	— Doctorate	需先修計算機網路 prerequisite: Computer Networks
219165	弗列斯科網頁程式設計 Flask: Web Development with Python	3	選 (E)	— Doctorate	需先修Python程式設計 prerequisite: Python Programming
219166	方程式最佳化 Function Optimization	3	選 (E)	— Doctorate	需先修高等計算智能 prerequisite: Advanced Computational Intelligence
219167	即時串流協議 Real-Time Streaming Protocol	3	選 (E)	— Doctorate	需先修計算機網路 or 電腦網路通信協定 prerequisite: Computer Networks or Computer Network Protocols
219169	資訊科學特論 Special Topics in Computer Science	3	選 (E)	— Doctorate	
219168	物聯網晶片開發實務 Practices of Internet of Things Chip Development	3	選 (E)	— Doctorate	需先修程式設計、邏輯設計與實驗(一)、邏輯設計與實驗(二) prerequisite: Programming、Logic Design and Laboratory Experiments (1)、Logic Design and Laboratory Experiments (2)
219170	K8S 微服務管理入門 Managing Microservices using Kubernetes	3	選 (E)	— Doctorate	
219171	深度學習實驗 Deep Learning Lab	3	選 (E)	— Doctorate	
219172	C語言安全程式設計 Secure Coding in C and C++	3	選 (E)	— Doctorate	需先修程式設計 prerequisite: Programming
219173	MPI平行程式設計 Parallel Programming with the Message-Passing Interface	3	選 (E)	— Doctorate	需先修程式設計, 計算機網路 prerequisite: Programming, Computer Networks

219174	秘密分享原理與應用 The Principle and Application of Secret Sharing	3	選 (E)	— Doctorate	需先修離散數學 prerequisite: Discrete Mathematics
219175	區塊鏈原理與應用 Principles and Practice of Blockchains	3	選 (E)	— Doctorate	
219176	網路攻防技術 Network Attack and Defense	3	選 (E)	— Doctorate	
219177	Golang程式設計 Golang Programming	3	選 (E)	— Doctorate	
219178	Python 開發維運一體化 Python for DevOps	3	選 (E)	— Doctorate	
219179	資料中心管理實務 Data Center Management Practices	3	選 (E)	— Doctorate	
219180	機器學習的數學方法 Mathematical method for machine learning	3	選 (E)	— Doctorate	
219181	人工智慧導論 Introduction to Artificial Intelligence	3	選 (E)	— Doctorate	
219182	電腦視覺與人機介面 Computer Vision and Man-machine Interface	3	選 (E)	— Doctorate	
219183	資料壓縮 Data Compression	3	選 (E)	— Doctorate	
219184	電腦圖學 Computer Graphics	3	選 (E)	— Doctorate	
219185	Web技術 WEB Technologies	3	選 (E)	— Doctorate	
219186	高等演算法（一） Advanced Algorithms (I)	3	選 (E)	— Doctorate	
219187	高等演算法（二） Advanced Algorithms (II)	3	選 (E)	— Doctorate	
219188	虛擬實境 Virtual Reality	3	選 (E)	— Doctorate	
219189	多媒體通訊 Multimedia and Communication	3	選 (E)	— Doctorate	
219190	高等計算機結構 Advanced Computer Architecture	3	選 (E)	— Doctorate	
219191	生物計算 Computational Biology	3	選 (E)	— Doctorate	
219192	自然語言處理 Natural Language Processing (NLP)	3	選 (E)	— Doctorate	
219193	資料庫系統 Database System	3	選 (E)	— Doctorate	
219194	高等電腦視覺 Advanced Computer Vision	3	選 (E)	— Doctorate	
219195	高等電腦圖學 Advanced Computer Graphics	3	選 (E)	— Doctorate	
219196	計算理論 Theory of Computation	3	選 (E)	— Doctorate	

219197	平行計算 Introduction to Parallel Algorithms	3	選 (E)	— Doctorate	
219198	資訊檢索 Information Retrieval	3	選 (E)	— Doctorate	
219199	通訊導論 Introduction to Communications	3	選 (E)	— Doctorate	
219200	圖論演算法 Graph Algorithms	3	選 (E)	— Doctorate	
219201	完美圖理論 Perfect Graph Theory	3	選 (E)	— Doctorate	
219202	錯誤更正碼 Error Correcting Code	3	選 (E)	— Doctorate	
219203	編碼理論 Coding Theory	3	選 (E)	— Doctorate	
219204	圖形理論 Graph Theory	3	選 (E)	— Doctorate	